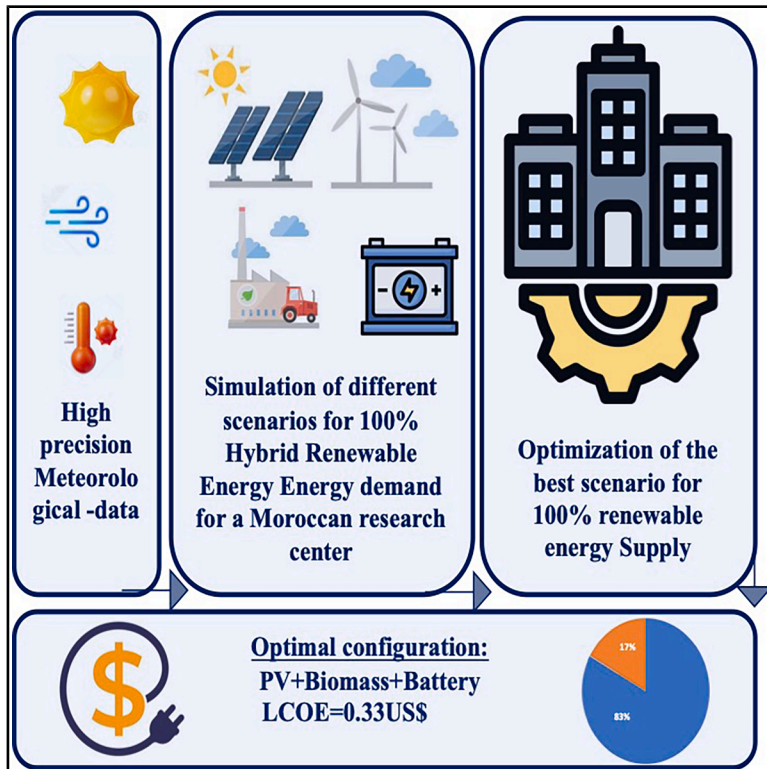


Techno-economic analysis for a 100% renewable hybrid energy systems integration in a research facility in Morocco

Graphical abstract



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In brief

Energy systems; Electrical Engineering; Energy Sustainability

Highlights

- Assessing different Renewable energy combinations for a 100% powering a research center
- High quality ground measurement data have been used to enhance the simulation results
- PV/Biomass/Battery with 83.8% solar energy share is the optimal combination
- The optimal configuration provides the lowest (LCOE) at \$0.33/kWh with 35% energy surplus



Article

Techno-economic analysis for a 100% renewable hybrid energy systems integration in a research facility in Morocco

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SUMMARY

The objective of this study is to assess the optimal design of hybrid renewable energy systems (HRES) to achieve a 100% energy supply for a research institute located in mid-south Morocco. Using high-precision meteorological data and HOMER software simulations, and optimizing of various combinations of photovoltaic (PV) solar panels, wind turbines, biomass generators, and battery storage systems were done. Nine different scenarios were analyzed to identify the most cost-effective and technically viable system to meet the institute's energy demands. Results show that the scenario including PV/biomass/battery combination provides the optimal solution to meet 100% energy demand from both technical and economic point of view with an LCOE of 0.33US\$ and a total share of solar energy of 83.3%. These results are of significant importance as they offer valuable insights into the most suitable technologies and combinations for achieving a 100% green energy supply to Moroccan governmental organizations.

INTRODUCTION

Climate change continues to be a persistent issue worldwide, as approximately 74% of global energy needs are met by fossil fuels, which significantly contribute to this ongoing predicament due to their greenhouse gas emissions.¹ Moreover, energy demand is expected to rise continuously because of the essential role of energy in the current economy and population growth, which is forecasted to reach 8.1 billion by 2025.² To achieve the climate targets of the Paris Agreement, greenhouse gas emissions must be reduced.³ Therefore, the integration of renewable energy sources into the current energy system is globally promoted to satisfy world energy needs while simultaneously protecting the environment.^{4,5} In fact, it is projected that the energy supply from renewable sources will range between 43% and 67% by 2050. However, further growth in the share of renewable energy in total primary energy is required, increasing from 15% in 2015 to 60% by 2030, and reaching 85% by 2050.⁶

Indeed, the current development of renewable energy technologies has led to a decrease in their costs. With the availability of various renewable energy sources, such as solar and wind, power generation from these sources can be a sustainable and feasible option from both technical and economic perspectives.^{7,8} Additionally, these technologies offer flexibility in deployment across different scales-large, medium, and small-as well as in standalone applications, enabling the provision of

a 100% renewable energy system that meets the load demand of various microgrids in sectors such as residential, industrial, and public facilities.⁹

However, the main drawback of renewable energy-based technologies is their intermittent power production, as they depend on meteorological conditions.¹⁰ For instance, the performance of photovoltaic (PV) technology is highly influenced by solar radiation and ambient temperature, and it cannot generate electricity during nighttime. In contrast, wind turbine production is directly related to wind velocity.¹¹ To address the issue of stochastic production, wind and PV technologies can be combined in what is known as a hybrid energy system, which aims to overcome the limitations of each technology and increase overall system efficiency.¹² Nonetheless, a storage medium is required for the hybrid energy system to ensure stable and reliable power generation that meets the load demand of a microgrid.¹³ Options include batteries that can store excess energy for use during production shortages, or biomass-based generators that, in combination with solar and wind, can provide clean and continuous power production.^{14,15}

The main challenge regarding hybrid energy systems is to identify the most optimal combination of the various system components from both technical and economic standpoints. In fact, the costs of the hybrid system are determined by the costs associated with each technology (capital, operation and maintenance, replacement, and so forth) and the overall system size needed to meet load demand. Therefore, the key to an optimal



hybrid energy system is to develop the most cost-effective design to provide the required energy.¹⁶

For the simulation, design, and optimization of hybrid energy systems, various tools and models are available, with one of the most powerful and widely used being the Hybrid Optimization Model for Electric Renewables (HOMER), developed by the National Renewable Energy Laboratory (NREL).^{15,17} Many studies in the literature have employed HOMER to evaluate and optimize hybrid energy systems in different locations. For instance, Li et al.¹⁸ assessed the use of renewable energy sources for energy production in a small village located in Nanyang, Henan Province, China. To find the optimal configuration, the authors utilized HOMER Pro software and determined that the most economically viable combination includes a 250 kW PV plant, a 360 kW diesel generator, a 100 kW wind turbine, a 150 kW converter, and 10 battery units. The cost of energy production from this configuration (COE) is 0.378 \$/kWh with a net present cost (NPC) of 1,795,558 \$. Similarly, Kapen et al.¹⁹ conducted a feasibility study of two hybrid energy systems for power and hydrogen generation in Maroua, Cameroon. The first system is composed of PV, fuel cell, electrolyzer, and biogas, whereas the second consists of PV, battery, fuel cell, electrolyzer, and biogas, and both systems were optimized for 3 electricity demand profiles: low, medium, and high. The results show that, the use of batteries in the second scenario lead to a reduction in the levelized cost of electricity (LCOE) and hydrogen (LCOH) as the optimal LCOE for the first configuration were found equal to: 0.871 \$/kWh, 0.898 \$/kWh and 1.524 \$/kWh, and for the second configuration: 0.139 \$/kWh, 0.091 \$/kWh and 0.071\$/kWh for the low, medium and high profiles, respectively. While for the LCOH it was found for the first configuration: 7.66 \$/kg, 4.95 \$/kg and 0.45 \$/kg and for the second configuration: 3.06 \$/kg, 1.34 \$/kg and 0.15 \$/kg for the low, medium, and high profiles, respectively. Turkdogan²⁰ design and optimized a 100% renewable hybrid energy system composed of wind turbines, PV panels, electrolyzer, hydrogen tank and battery to meet the energy needs of a single-family house for both electricity and transportation in the Ayvalik province, Turkey. It was found that the optimal scenario leads to an LCOE of 0.685 \$/kWh and an LCOH of 6.85 \$/kg. In addition, the sensitivity analysis results show that a reduction of 50% in the component costs leads to 26.4% decrease of the LCOE and LCOH. On the same vein, Chaichan et al.²¹ investigated the use of a hybrid energy system for the power generation of green islands in Thailand. The authors evaluated four scenarios that incorporate solar photovoltaic, wind energy, fuel cell and batteries with the use of diesel generation or grid connection to the mainland as backup and the island of Koh Samui in southern Thailand has been used as a case study. The results found that the lowest LCOE was achieved by a system composed by the hybrid renewable energy system coupled with the grid with 0.132 \$/kWh, however, this scenario leads to high greenhouse gas emissions costs due to Thailand's carbon-intensive power portfolio. Accordingly, a 100% renewable standalone hybrid energy has been considered as more suitable for the Koh Samui island to satisfy its load demand with an LCOE of 0.309 \$/kWh. Odetoye et al.²² evaluated the use of standalone renewable energy microgrid in a growing commercializing community in Nigeria's middle belt. To do so,

the authors considered a mix of different renewable energy systems composed of photovoltaic, concentrated solar thermal, hydropower, and batteries. The optimal configuration of the renewable energy system was found to be more economically and environmentally viable in comparison to diesel generators typically used for similar applications. In fact, The Net Present Cost was found to be 55.7 million \$ and 408 million \$ while the LCOE was equal to 0.26 \$/kWh and 1.01 \$/kWh for the renewable and diesel based energy system respectively. Chisale et al.²³ tackled the electricity power shortage problem in Malawi, especially for schools. Accordingly, six scenarios of hybrid energy systems were evaluated from a technical and economic perspective that include the grid, diesel generator, solar photovoltaic, wind energy, biomass, and batteries, while the simulations were conducted under the conditions of Blantyre secondary school. As a result, a combination of solar photovoltaic, the grid and biomass was identified as the most optimal configuration that leads to an LCOE of 0.095 \$/kWh, which is lower than the Malawian grid electricity cost (0.11 \$/kWh) as well as reducing the greenhouse gas emissions.

Similarly, Baneshi and Hadianfard²⁴ designed a hybrid energy system composed of diesel generator, solar photovoltaic, wind energy and batteries with the aim to satisfy the power requirements of 9.911 MWh daily average with a peak of 725 kW for a large electricity consumer located in Shiraz southern Iran. The study evaluated two scenarios on and off grid connection and the results found that the LCOE varies between 5.7 and 8.4 €/kWh with the variation of renewable energy fraction (solar and wind) between 0 and 53% for the grid connected system, whereas for the off grid system the LCOE is in the range of 9.3–12.6 €/kWh as the renewable energy fraction varies between 0 and 43.9%. Pal and Mukherjee¹³ proposed an optimal design of a hybrid standalone energy system composed of solar photovoltaic and hydrogen based fuel cell for renewable electricity supply for the north-eastern states of India. The results show that for an average daily demand of 138 kWh the optimal configuration based on the lowest net present cost and LCOE is in the range of 110–120 kW for the photovoltaic plant, 40–60 kg for the hydrogen tank, 30–60 kW for the electrolyzer and 15 kW fuel cell with an LCOE between 0.509\$ and 0.689 \$/kWh and net present cost between 327,557 and 443,004 \$. Li et al.²⁵ conducted a techno-economic feasibility study on a hybrid system composed of diesel generator, wind energy and batteries for power generation for 280 residential houses situated in Lanzhou suburbs at Gansu province in China. The results found that the optimal configuration were find in the case of the system composed by diesel generator and Zinc-Bromine batteries with a net present cost and an LCOE of 7,735,646\$ and 0.471\$/kWh respectively.

This study focuses on determining the optimal design and performance of hybrid renewable energy systems to fully meet the energy needs of the Green Energy Park, a research institute in Benguerir, Morocco. By leveraging detailed local meteorological data, the HOMER software was used to simulate and analyze various combinations of renewable technologies, including photovoltaic (PV) solar panels, wind turbines, biomass generators, and battery storage. A total of nine different configurations were tested, with the goal of identifying the most economically

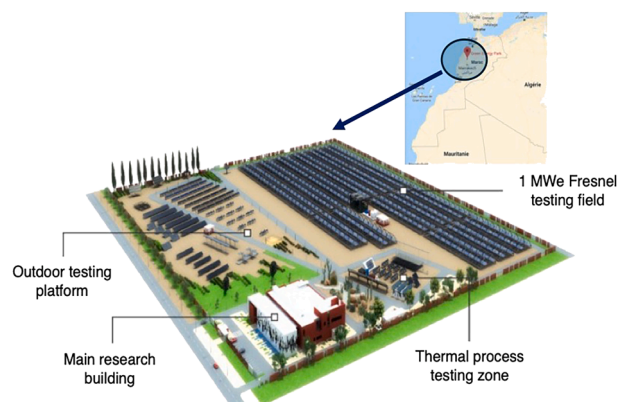


Figure 1. Green Energy Park research facility

efficient and technically robust system capable of supplying 100% of the institute's power demands.

RESULTS

The objective of this section is to present the field of study, including the electrical consumption, the meteorological conditions as well as the description simulation tools, mathematical equations, and simulation inputs.

Site description and electricity load curve

The study is conducted at Green Energy Park research (GEP) facility, located in Benguerir (Mid-south of Morocco, see Figure 1). This research facility is the first in Africa presenting a cutting edge platform for testing different solar technologies under real conditions of a semi-arid climate. GEP is composed of an outdoor testing field and a building including offices and laboratories. These laboratories are equipped with several equipment's that consumes massively the electricity, such as the scanning electronic microscope (S.E.M) and the Plasma Enhanced CVD Furnace (PECVD) for nano-coatings deposition and cells developments. In addition to that, GEP is equipped with PV modules IEC testing performances such as the electroluminescence, flash test, the mass load, and a small lamination platform for building small and medium PV modules. For more details regarding the GEP platform.

Regarding the labor, GEP has 22 permanent scientists working the from 9:00 a.m. to 5:00 p.m. from Monday to Friday, nevertheless, this number increases significantly between March and July where the GEP research center received around 200 students to conduct their internship and research projects, this increases massively the electricity consumption -reaching 120 kW- during this period due to the use of the majority of the equipment to conduct the experiments as well as the use of the air conditioning systems since this period is characterized with high temperature records. The hourly electricity load measured at 1 h step is presented in Figure 2.

Renewable resources assessment

To evaluate the performance of renewable energy integration for a location or a building, one of the most important parameters that has to be assessed is the potential of the renewable resources such as solar and wind. For a high accuracy evaluation and realistic simulation outputs, these parameters should be measured with high accuracy instruments characterized by low measurement error. Accordingly, in this study, the simulation inputs were extracted from *in-situ* measurements using a high precision meteorological station installed at GEP, see Figure 3.

This station used is fully automated and equipped with a sun tracker capable of measuring the three components of solar irradiance separately and simultaneously, along with standard meteorological data. The global horizontal irradiance (GHI) and diffuse horizontal irradiance (DHI) components were measured using the Kipp & Zonen CMP21 Class A pyranometer, which has a spectrally flat ISO 9060 response and an accuracy of $\pm 2\%$. It is worth mentioning that for the DHI measurements, the pyranometer is equipped with a shadow-ball that blocks the direct component of the solar irradiance.

As for the direct component (DNI), it is measured using an ISO 9060 spectrally flat class A CHP1 pyrheliometer with an error lower than 2%. Ambient temperature and humidity were recorded using the Campbell CS215 sensor, with accuracies of $\pm 0.3^\circ\text{C}$ and $\pm 2\%$, respectively. As for Wind speed and direction, they were measured using the NRG 40H anemometer and NRG #200P wind vane, both with errors under 1%.

The measurements were recorded at 10-s intervals using the Campbell Scientific R1000 datalogger, and they were converted to 1 h data -after a thorough data analysis and quality control process-to be used for the simulations. We need to mention

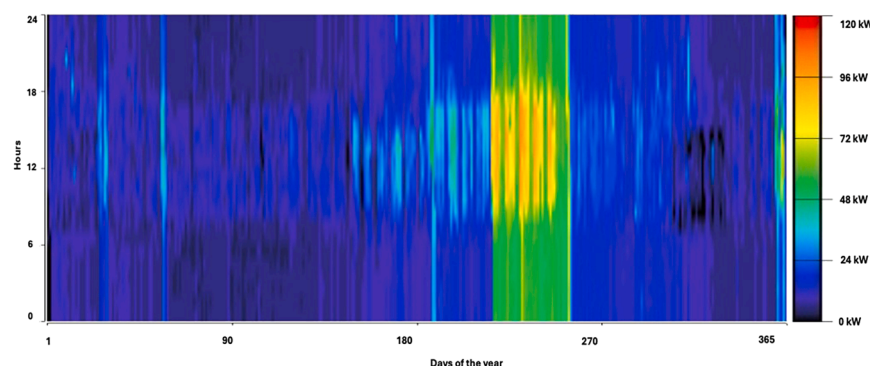


Figure 2. The hourly electricity load curve measured at Green Energy Park

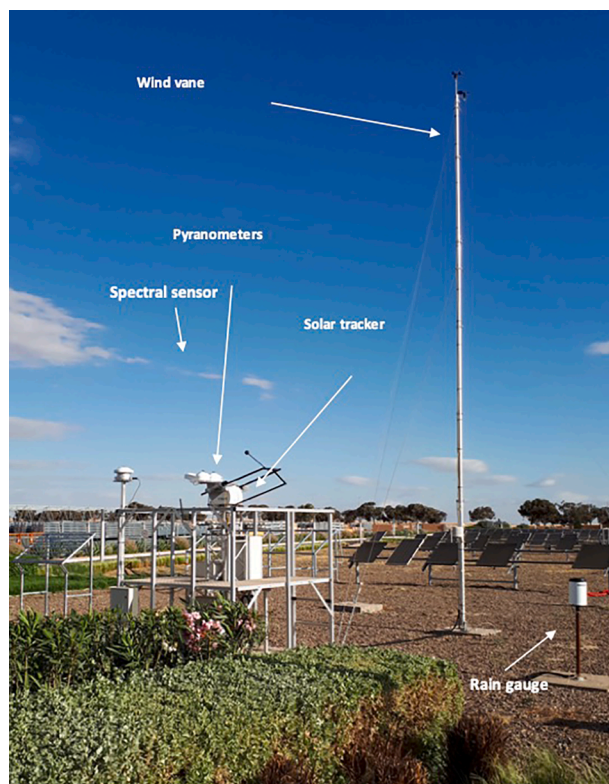


Figure 3. Meteorological station at Green Energy Park

that all sensors were carefully maintained and regularly calibrated to guarantee high-quality data.

Assessment of solar energy resources

The performance of PV modules is greatly influenced by the amount of solar radiation received, primarily the global horizontal irradiance (GHI), as well as ambient temperature. The monthly average variations of GHI and ambient temperature at the GEP site are shown in the Figure 4. The data reveal that the study area experiences high levels of solar irradiance, particularly between May and August, when daily average irradiance can reach up to 341 W/m². Even during fall and winter, GHI values remain significant (around 150 W/m²), which is relatively high compared to neighboring countries such as Spain and Portugal.

As for temperature, we observe moderate values during fall and winter, averaging around 10°C. However, in summer, temperatures rise significantly, reaching up to 28°C, which can negatively impact the performance of PV modules.

Assessment of wind resources

For the assessment of wind resources, given that wind speed and direction frequently fluctuate due to geographical factors and atmospheric circulation,²⁶ evaluating the frequency of wind speed and direction is more appropriate. Accordingly, the wind rose diagram, presented in the Figure 5, offers a clear visualization of both parameters in a single plot. As shown, the most frequent wind direction is from the west, while wind speeds predominantly remain below 5 m/s; in fact, the most

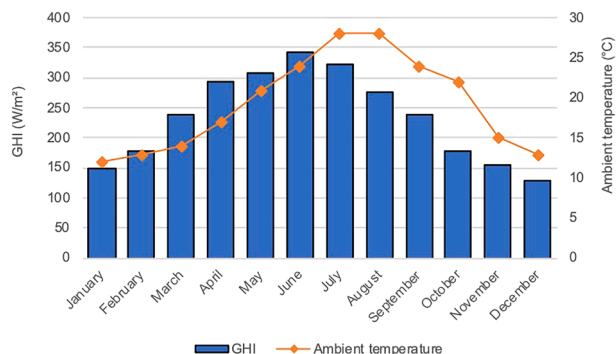


Figure 4. The monthly average variation of GHI and ambient temperature

frequent wind speed range is between 0 and 3.5 m/s. Indeed, according to the National Renewable Energy Laboratory (NREL), depending on wind speed, the location can be classified into a wind power class. The most suitable locations are classified in class 4 and higher, whereas class 3 is suitable for high hub wind turbines, while class 2 requires an appropriate consideration. As for class 1, they are not suitable for any wind energy based project.²⁷

The annual average wind speed in the Benguerir region is 2.89 m/s at 10 m, which means that it belongs to class 1. This suggests that the electricity production from the wind turbine will be low, leading to higher costs.

Simulation tool and hybrid energy systems design specifications

HOMER is a powerful simulation tool for the design, simulation, optimization, and economic analysis of hybrid energy systems.²⁸ Indeed, even though many simulation software exist for hybrid renewable energy system simulation, HOMER is the most used tool for the techno-economic analysis, optimization, and design of hybrid renewable energy systems, due to its flexibility, fast simulation and optimization process, and more importantly, accuracy.¹⁵ Its key feature is the ability to identify the most cost-effective and technically feasible design from various combinations of power generation systems to meet a specific load demand.²⁹ To achieve this, HOMER requires input data such as meteorological information (GHI, ambient temperature, and wind speed), load profiles, system component characteristics, and technical and economic data.¹⁷

During the simulation and optimization process, HOMER evaluates different system configurations to meet the load demand while accounting for the technical limitations of each component. Each configuration is modeled over a specified period, and an optimization algorithm is used to identify the optimal solution that minimizes both the Net Present Cost (NPC) and Levelized Cost of Energy (COE) from the set of optimized configurations.^{30–33} The total NPC is calculated using the following equation³⁴:

$$C_{NPC} = \frac{C_{ann,tot}}{CRF(i, R_{proj})}$$

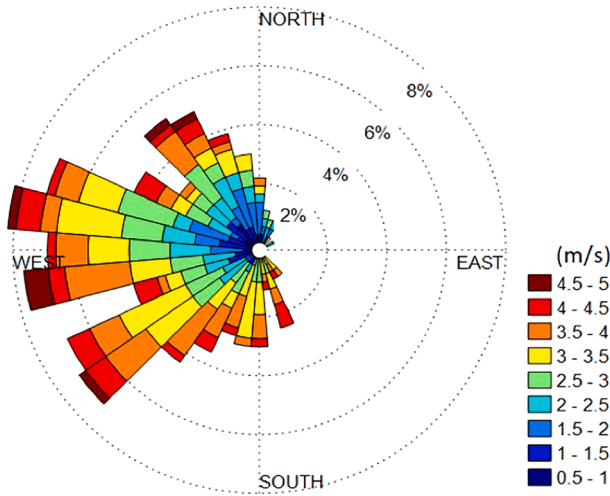


Figure 5. Wind rose for Green Energy Park research facility

Where $C_{ann,tot}$ is the total annualized cost (\$/year), CRF is the Capital Recovery Factor, i is the real interest rate, and R_{proj} is the project lifetime (years). The CRF , is calculated as follows³⁵:

$$CRF(i, N) = \frac{i(1+i)^N}{(1+i)^N - 1}$$

Where N is the number of years.

Regarding the levelized cost of energy (COE), its calculated with the following equation³⁶:

$$COE = \frac{C_{ann,tot}}{E_{ser}}$$

$C_{ann,tot}$ represents the total annualized cost of the system (\$/year), and E_{ser} is the total electrical load served (kWh/year).

In this study, the project lifetime is 20 years, while the inflation rate is 2.5%, and the discount rate is 5%.^{37,38}

Hybrid energy system components and costs

In this part, we will present the technical specification of each component of the hybrid energy system along with its corresponding costs as well as an overview of the model used by HOMER to calculate its power generation.

PV module

The PV array simulated in this study consists of fixed PV panels using polycrystalline silicon solar cell technology. The power generated by PV modules is calculated using the following formula³⁹:

$$P_{PV} = Y_{PV} \cdot f_{PV} \cdot \left(\frac{G_{PV}}{G_{ref}} \right) \cdot [1 + K_T(T_C - T_{ref})]$$

Where Y_{PV} represents the PV generation under standard test conditions (kW), Y_{PV} is the PV nominal capacity and f_{PV} the de-

Table 1. Characteristics of the PV module

Properties	Specifications
Brand name	Generic flat plate PV
Rated capacity (kW)	1
Temperature Coefficient	-0.42
Operating Temperature (°C)	44.80
Efficiency (%)	14.40
lifetime	25 years

rating factor. G_{PV} solar radiation received by the PV array and G_{ref} represent the incident solar energy on the PV array under standard test condition. K_T is the temperature coefficient of the maximum power and T_C and T_{ref} are the temperature of the PV solar cell temperature and PV cell temperature under at standard test conditions, respectively. According to the Moroccan market, the capital cost of the PV system is taken as at 1600 \$/kW. While the operation and maintenance cost is set to 15 \$/year.³⁸ The technical characteristics of the PV panel are presented in Table 1.

Wind turbine

The mechanical energy created by the movement of the turbine blades under the influence of wind speed is converted into electricity via a generator. The power output is calculated using the following equation⁴⁰:

$$P_{Wout} = \frac{1}{2} \cdot \rho \cdot A \cdot V \cdot C_p(\alpha, \beta) \cdot \eta_t \cdot \eta_g$$

Where ρ is the air density, A is the rotor area in m^2 , V is wind velocity in m/s, C_p is the performance density of the air coefficient of the turbine and η_t and η_g are the efficiencies of the wind turbine and the generator, respectively. Regarding system costs, the capital cost is 10775 \$, the operation and maintenance costs is \$20 per year.³⁸ With respect to the replacement cost for both PV and wind turbines, it is not considered since the lifetime of both technologies is equal to the project life span. The technical details of the selected wind turbine in this study are presented in Table 2.

Battery system

The incorporation of batteries in standalone hybrid energy systems is essential due to the intermittent nature of renewable energy sources. Batteries help maintain a stable electrical supply, especially during peak demand hours. In this study, we used a 1 kWh Li-Ion battery with a capacity of 167 Ah and a nominal voltage of 6V. The technical specifications of the battery are summarized in Table 3. In the Moroccan market, the capital and replacement cost of this battery is approximately 330 \$, with an annual maintenance cost of 10 \$. The battery is expected to have a lifespan of 15 years.³⁸

Converter

Converters are a crucial component of hybrid energy systems, as they enable the conversion between direct current (DC) and alternating current (AC). This is necessary because some equipment, such as batteries and PV modules, operates on DC, while others, such as wind turbines, generate AC.⁷ In this study, the inverter efficiency is set at 90%, while the rectifier efficiency is

Table 2. Characteristics of the Wind turbine

Properties	Specifications
Brand name	AWS HC 5.1 kW Wind turbine
Rated capacity (kW)	5.1
Number of blades	3
Rotor diameter (m)	5.24
Hub height (m)	12
lifetime	25 years

95%. The capital and replacement costs of the converter are both 400 \$, with no annual maintenance costs.³⁸

Biogas generator

The use of biomass energy is a favorable way to generate renewable electricity; however, the biomass source should be carefully chosen.⁴¹ In this study, olive waste residues are used as a biomass source due to their abundance in Morocco in general as the country is Morocco is ranked in the fifth position worldwide in terms of production and export, with an annual average production of 1.92 million tons⁴² and the Benguerir region in particular, to avoid any shortage in terms of biomass feedstock supply. The biomass has a lower heating value of 18.750 MJ/kg, and its cost is \$60 per ton.^{43,44} The biogas generator considered has an operational lifespan of 20,000 h, with a capital cost of 1,600 \$ per kW. The replacement cost is 1,250 \$ per kW, and the operation and maintenance (O&M) costs are 0.10 \$ per operational hour.³⁸

DISCUSSION

As previously mentioned, HOMER was used to simulate nine scenarios incorporating various combinations of different technologies. The aim was to identify the most optimal combination of available renewable energy sources. It is important to note that all proposed scenarios ensure 100% satisfaction of the GEP load demand, with no capacity shortages. The following Renewable Energy Hybrid Systems (HRES) were simulated in this study (see Figure 6).

- (1) Combination 1: PV/Wind/Biomass
- (2) Combination 2: PV/Battery
- (3) Combination 3: PV/Biomass
- (4) Combination 4: PV/Biomass/Battery
- (5) Combination 5: PV/CSP/Biomass/Wind/Battery
- (6) Combination 6: PV/Wind/Battery
- (7) Combination 7: PV/Wind/Biomass/Battery
- (8) Combination 8: Wind/Biomass
- (9) Combination 9: Wind/Biomass/Battery

HOMER simulation result

In the optimization process using HOMER, multiple scenarios were generated for each combination. However, in this study, we focus on the optimal scenario for each combination. The results of these optimal scenarios are discussed later in discussion.

Combination 1: PV/wind/biomass

In this scenario, as illustrated in Figure 6, a combination of photovoltaic (PV) solar panels, biomass, and wind turbines is employed. The system capacities, as shown in Table 4, are

Table 3. Battery characteristics

Properties	Specifications
Nominal capacity (kWh)	1 kWh
Nominal capacity (Ah)	167 Ah
Nominal voltage	7 V
Maximum discharge current	500 A
Maximum charge current	167 A
Lifetime	15 years

distributed as follows: 110 kW for PV solar panels, 75 kW for biomass, 23 wind turbine units, and 71 kW converter capacity. The estimated annual energy production is 364.669 MWh, with a surplus of 51.6%, corresponding to an excess of 188 MWh. The Levelized Cost of Energy (COE) is calculated at 0.668 \$/kWh.

The energy outputs from wind, PV, and biomass are 66.009 MWh/year, 235.040 MWh/year, and 63.620 MWh/year, respectively. These figures highlight the contribution of each technology, with PV being the largest contributor at 64.5%, followed by wind energy at 18.1%, and biomass at 17.4%. This dominance of PV is attributed to the significant solar energy potential at the study site, resulting in high energy production at a lower cost.

Combination 2: PV/battery

In this configuration, photovoltaic (PV) solar panels and battery systems are utilized, as illustrated in Figure 6. The system capacities are detailed in Table 5, with 352 kW allocated for the PV solar panels, 1,437 kWh for the battery systems, and a converter capacity of 122 kW. This results in a Levelized Cost of Energy (COE) of 0.555 \$/kWh. The total energy production is estimated at 751 MWh per year, with 81.814 MWh/year supplied from the battery. The energy surplus represents 75% of the total production, equating to an excess of 562.9 MWh.

This high energy surplus is primarily due to the PV array being the sole power generation source. It is sized to meet load demands, including peak hours, meaning that during periods of low electricity demand or when battery storage is not required, the PV system continues to produce energy, leading to significant excess.

Combination 3: PV/biomass

The system comprises the components shown in Figure 6, including a photovoltaic (PV) array and a biomass system. As detailed in Table 6, the system specifications are as follows: a 122 kW PV module and a 75 kW biomass system, along with a converter capacity of 72 kW. The estimated annual energy production is 348.802 MWh, with an available energy surplus of 48.6%, corresponding to an annual energy excess of 169.68 MWh.

In this configuration, the contribution from PV significantly predominates, accounting for 74.6% of the total energy production, while biomass contributes only 25.4%. This distribution is logical, given that the cost of PV is substantially lower than that of biomass, particularly in light of the high solar resources available at the study site.

Combination 4: PV/biomass/battery

In this configuration, the system combines photovoltaic (PV) solar panels, biomass, and a battery system, as illustrated in

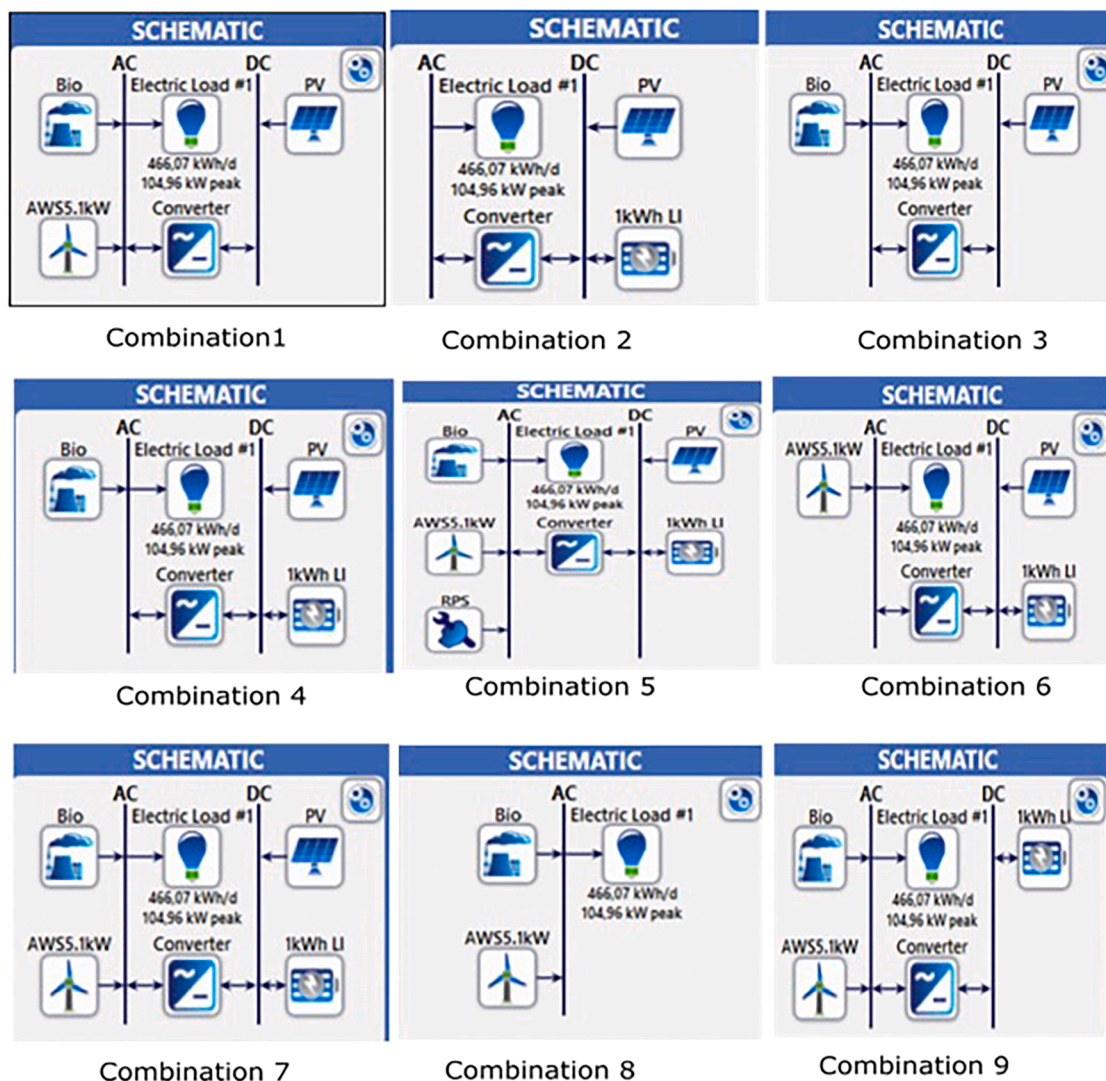


Figure 6. Proposed HRES for application in the study site

Figure 6. The optimal results are presented in Table 7, showing capacities of 117 kW for the PV, 75 kW for the biomass, and 371 kWh for the battery systems, along with a 104 kW converter

Table 4. PV/wind/biomass technical details

Architecture	PV (kW)	110
	Wind (kW)	117.3
	Biomass (kW)	75
Energy Production (MWh/years)	PV	235.040
	Wind	66.009
	Biomass	63.620
Energy Contribution %	PV	64.5
	Wind	18.1
	Biomass	17.4
COE (\$/kWh)	–	0.668

capacity. The annual energy production for this configuration is estimated at 298.883 MWh, resulting in an energy surplus of 35.5%, which corresponds to a yearly excess of 106.13 MWh.

The incorporation of batteries has significantly reduced expenses, lowering costs by more than half compared to the configuration without batteries (Combination 3). Additionally, this modification has slightly decreased the contribution share of biomass compared to the previous configuration.

Combination 5: PV/CSP/biomass/wind/battery

In this combination, all the possible components available at the GEP, namely, photovoltaic solar panels (PV), Concentrated Solar Power (CSP), biomass, wind turbines, and battery systems, were combined, as illustrated in Figure 6. The results from the HOMER software suggest that the optimal configuration in this case is the one without considering CSP integration. This configuration is already emphasized in combination 4, therefore, this scenario will not be considered in the analysis of this study. Furthermore,

Table 5. PV/battery

Architecture	PV (kW)	352
Energy Production (MWh/years)	PV	751.004
Energy Contribution %	PV	100
Battery	batteries (kWh)	1437
COE (\$/kWh)	–	0.555

several simulations were conducted for scenarios using CSP with other renewable energy technologies, and none of the optimization results selected the CSP technology as a part of the HRES design. This is due to the high costs associated with the CSP technology since it is more suited for large-scale applications rather than for microgrid applications.

Combination 6: PV/wind/battery

In this scenario, a combination of photovoltaic (PV) solar panels, wind turbines, and battery systems is employed, as illustrated in Figure 6. The system capacities are as follows: 351 kW for PV solar panels, 8 wind turbines corresponding to a capacity of 40.8 kW, and 1,275 kWh for the battery systems (refer to Table 8), along with a converter capacity of 110 kW. The estimated annual energy production from this system reaches 773.58 MWh, with an energy surplus of 73.8%, representing 579 MWh.

In this configuration, PV makes a significant contribution of 97%, while wind energy contributes only 3%, translating to an annual production of 750.62 MWh from the PV system and 23 MWh from the wind energy system. This disparity is attributed to the limited wind resource available at the study site compared to solar irradiation, as noted in the resource assessment section. In fact, the wind turbines primarily generate electricity during nighttime.

Regarding the use of batteries, it is observed that their nominal capacity has decreased compared to Combination 2 (PV/Battery) due to the addition of wind turbines, which produce 74.41 MWh of electricity during the night. This reduces reliance on the battery system alone. The battery output in this configuration is therefore 74.41 MWh.

Combination 7: PV/wind/biomass/battery

Similar to the fifth combination, HOMER did not propose any optimal system that includes all components for this configuration, aside from those already evaluated in the previous combination. Therefore, this combination will also be excluded from the analysis of this study.

Combination 8: Wind/biomass

In this combination, biomass and wind turbines are integrated, as illustrated in Figure 6. The simulation results, presented in

Table 6. PV/biomass

Architecture	PV (kW)	122
	Biomass (kW)	75
Energy Production (MWh/years)	PV	260.254
	Biomass	88.548
Energy Contribution %	PV	74.6
	Biomass	25.4
COE (\$/kWh)	–	0.708

Table 7. PV/biomass/battery

Architecture	PV (kW)	117
	Biomass (kW)	75
Energy Production (MWh/years)	PV	250.478
	Biomass	48.405
Energy Contribution %	PV	83.8
	Biomass	16.2
Battery	batteries (kWh)	371
COE (\$/kWh)	–	0.333

Table 9, indicate capacities of 100 kW for biomass and 64 wind turbines, which collectively represent a rated capacity of 326 kW. Notably, no converter is used in this configuration, as both the wind turbine and biomass generator produce AC current.

This configuration achieves an annual energy production of 273.439 MWh, with an energy surplus of 37.8%, corresponding to an energy excess of 103.32 MWh. Remarkably, wind energy dominates the production, accounting for 67.2% of the total, while biomass contributes only 32.8%. However, the Levelized Cost of Energy (COE) for this configuration is 1.04 \$/kWh, which is significantly higher than that of the other configurations.

Combination 9: Wind/biomass/battery

The incorporation of battery systems has notably led to a substantial reduction in costs, nearly halving them when contrasted with the configuration lacking batteries (referred to as combination 8). Moreover, this adjustment has marginally increased the contribution of biomass compared to the previous combination.

In this setup, the hybrid energy system consists of wind turbines, a biomass generator, and a battery system, as illustrated in Figure 6. The simulation results are detailed in Table 10, indicating capacities of 75 kW for biomass, 33 wind turbines with a nominal capacity of 168 kW, 258 kWh for the battery systems, and a converter capacity of 61 kW. The annual energy production for this arrangement amounts to 203.692 MWh, accompanied by an energy surplus of only 7.9%, which represents an energy excess of 16.1 MWh. The incorporation of battery systems has significantly reduced costs, nearly halving them compared to the configuration without batteries (referred to as Combination 8). Moreover, this adjustment has slightly increased the contribution of biomass relative to the previous combination.

Optimization results comparison

In this section, we conduct a comparison of the various optimal hybrid energy system combinations presented above, focusing on the Levelized Cost of Energy (COE). Figure 7 illustrates the COE for each combination, ranked from highest to lowest.

The results show that the Wind/Biomass combination has the highest cost at 1.04 \$/kWh, followed by the PV/Biomass combination at 0.708 \$/kWh. The PV/Wind/Biomass combination has a COE of 0.668 \$/kWh, while the PV/Wind/Battery configuration is priced at 0.585 \$/kWh. The PV/Battery and Wind/Biomass/Battery combinations closely follow, with costs of 0.555 \$/kWh and 0.517 \$/kWh, respectively.

Overall, the PV/Biomass/Battery system emerges as the most advantageous option from both a technical and economic

Table 8. PV/wind/battery

	PV (kW)	351
Architecture	Wind (kW)	40.8
Energy Production (MWh/years)	PV	750.619
	Wind	22.960
Energy Contribution %	PV	97
	Wind	3
Battery	batteries (kWh)	1275
COE (\$/kWh)	–	0.585

standpoint, with the lowest COE of 0.333 \$/kWh, enabling the GEP Research Institute to operate independently from the grid. This cost is reasonable, especially considering that the hybrid energy system relies entirely on renewable energy sources. In contrast, the COE for the other combinations is significantly higher.

The following section will analyze various aspects of the PV/Biomass/Battery combination.

The relative contributions of each energy component are as follows: solar energy accounts for 83.8%, while biomass energy contributes 16.2%, as illustrated in Figure 8. As noted earlier, this scenario yields an energy surplus of 35.5%. This surplus energy can be utilized to charge the battery, allowing for storage and subsequent use during periods of high energy demand.

Figure 9 illustrates the average monthly variations in energy production from the PV/biomass/battery combination. It can be observed that the monthly electricity generation from the solar PV array remains relatively constant throughout the year, averaging around 21 MWh. Although the global horizontal irradiance (GHI) peaks in June at the study site, a slight decrease in electricity production from the solar panels in June and July can be attributed to the efficiency of the panels during these months. As previously mentioned, the study site is located in the Benguerir region, which experiences meteorological conditions such as sandstorms and strong winds.^{45–47} These conditions can lead to dust accumulation on the solar panels, acting as a thermal insulator that traps heat on their surface.^{48–52} This results in increased temperatures, compounded by the already high ambient temperatures during this period, as discussed in the solar resource assessment section. Consequently, this temperature rise can reduce the efficiency of the solar panels, significantly impacting their overall performance.

Electricity production from biomass remains low for most months of the year, particularly during March, April, May, and June, as the cost-effectiveness of electricity generated from PV is significantly higher than that from biomass. However, an exception occurs in August and September, when biomass energy production becomes notable, peaking in August. This increase is primarily due to the heightened electricity demand during this period, especially on late evenings -due to the presence of more people on the platform and the running experiments-necessitating more power to meet the energy needs. In August, biomass production exceeds that of PV, generating 26.2 MWh, while September also shows a substantial biomass output of 15.1 MWh.

Table 9. Wind/biomass

	Wind (kW)	326.4
Architecture	Biomass (kW)	100
Energy Production (MWh/years)	Wind	183.677
	Biomass	89.762
Energy Contribution %	Wind	67.2
	Biomass	32.8
COE (\$/kWh)	–	1.04

To better understand the power generation behavior of the various components throughout the year, we will present heat maps illustrating the hourly power production from the PV array and the biogas generator, along with the state of charge of the batteries, as shown in the respective data.

As shown in the Figure 10, the highest power production, approximately 90 kW, occurs mainly between 10:00 and 14:00, which aligns with the peak solar irradiation levels. It can also be observed that the PV array generates electricity efficiently for about 8 h daily throughout the year. However, during the summer months, production decreases compared to the rest of the year, despite the availability of high solar irradiation. This reduction is attributed to the impact of elevated ambient temperatures during this period, which negatively affects the performance of the PV solar cells, as mentioned earlier.

Regarding power production from the biogas generator, as shown in the Figure 11, it is primarily utilized during August and September, which correspond to the highest peak in load demand throughout the year. During the rest of the year, the PV array and batteries generate the necessary power, explaining the higher share of electricity produced by the PV system. Additionally, the biogas generator operates mainly at night and in the early morning, when solar irradiation is unavailable for the PV array. During these times, it works at full capacity (75 kW) to meet the load demand.

The battery performance is illustrated in the Figure 12, which provides a clear view of the energy flow in and out of the batteries. It can be observed that the period during which the batteries reach full charge (100%) occurs between 11:00 and 18:00, aligning with the peak production times of the PV system. This suggests that a significant portion of the energy generated by the PV array is directed toward storage. Subsequently, the state of charge of the batteries gradually decreases during the night and early morning, dropping to around 60% by the time the PV array begins generating high amounts of electricity again. This indicates that the stored energy in the batteries serves as the primary source for meeting load demand during these periods. The lowest state of charge is typically recorded during peak load demand periods, reaching around 20%. This coincides with the highest production from the biogas generator, allowing both components to work together to deliver the required power.

In terms of system economics, the associated costs for each component are presented in Table 11. As shown, the capital costs, replacement costs, operation and maintenance (O&M) costs, and total costs for each component of the optimal hybrid energy system are detailed. The respective total costs are \$471,515.59, \$114,210.80, and \$176,426.61. The highest cost

Table 10. Wind/biomass/battery

Architecture	Wind (kW)	168.3
	Biomass (kW)	75
Energy Production (MWh/year)	Wind	94.708
	Biomass	108.984
Energy Contribution %	Wind	46.5
	Biomass	53.5
Battery	batteries (kWh)	258
COE (\$/kWh)	–	0.517

contributor is the biogas generator, despite its lower capacity, which underscores the system's priority to utilize electricity from the PV array. Following the biogas generator are the PV array, batteries, and finally the converter. The overall Net Present Cost (NPC) of the system amounts to \$888,188.20.

In order to evaluate the environmental benefits of the PV/Biomass/Battery, the CO₂ emission reduction by the HRES is estimated. This done by calculating the avoided CO₂ when using the HRES in comparison to the case when the national grid is used to meet the research facility electricity demand. To do so, the following formula is used³⁸:

$$AV_{CO_2} = \frac{E_{PV}(f_{grid} - f_{PV}) + E_{biomass}(f_{grid} - f_{biomass})}{10^6}$$

Where AV_{CO_2} represents the annual mass avoided of CO₂ emissions in tones, E_{PV} and $E_{biomass}$ are the electricity production from the PV and biomass in (kWh) respectively. Whereas, f_{PV} is the life cycle emissions of the PV system taken as 50g of CO₂ for each kWh, and $f_{biomass}$ represent the carbon footprint of the biomass generation considered as 16.5g of CO₂ per kWh. As for f_{grid} is the electricity grid related carbon emission factor considered in Morocco equal to 746g CO₂/kWh.^{53,54}

As a result the potential CO₂ emissions reduction from adopting the HRES instead of the national electricity grid is estimated as 209.64 tones of CO₂ per year.

With the aim to benchmark this study findings, a comparative study is carried out with two similar studies in India since its considered as developing country as Morocco and therefore it can be considered to have similar economic and financial conditions as Morocco. The first one was conducted in Pirthala Village, Haryana, India.⁵⁵ This choice is due to many similar aspect with the current study: similar climatic conditions as the site is also located in arid region, HOMER was employed for design, simulation and optimization of an HRES to satisfy the village energy requirements from 100% renewable energy sources as well, and the site load profile has also a spike in energy consumption during summertime which is challenging for finding the HRES optimal design.

The site's main renewable energy technologies are solar PV and hydro power, with the use of batteries as a storage medium. The optimal configuration was found to be: 3461 kW for PV, 98.1 kW for hydro power, 2785 kW for the converter, and with 304 lithium-ion batteries, a storage capacity of 100 kWh leading to a COE of 0.782 US\$.

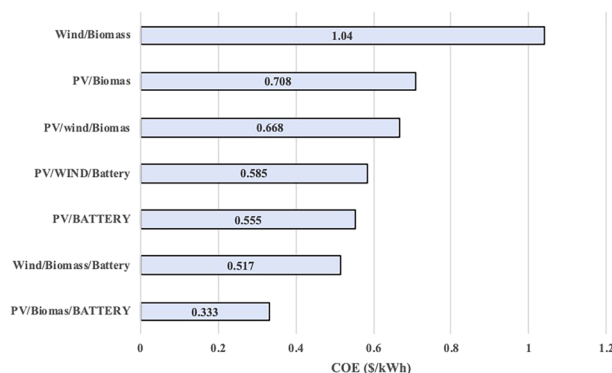


Figure 7. Comparison between the COE of the different optimal combinations

With respect to the proposed HRES in this study, the main difference in the architecture is the use of biomass instead of hydro power, which provides more reliability in terms of electricity generation due to the availability of the biomass source, whereas the hydro power generation depends on the stream flow of the hydro source. In fact, this reliability offers more flexibility for the HRES design as it can be used to help cover the electricity peak in the summer as observed in Figure 11, which leads to less battery capacity requirement and thus lower the system cost.

Indeed, the costs associated with the use of batteries can be considered as substantial, not only the capital cost but also, the O&M and the replacement costs, in fact, the costs associated with the batteries represent 73% of the overall costs of the HRES in Pirthala, while its only 24% in the GEP. This is one of the main causes of the difference in COE between the two HRES, since it is 0.782 US\$ for Pirthala and 0.333 US\$ for the GEP.

With respect to the second study, it was conducted in a rural village, Sidhwanbet, Punjab, India.⁵⁶ This study also used HOMER software for the design of a 100 renewable HRES similar to one in the current study composed since it's composed of solar PV, Biomass, and batteries. In addition, the location has an abundance of biomass sources from crop residue. However, the village load profile does not have a noticeable spike, as the electricity consumption is generally steady throughout the year. The optimal HRES design was found to be 1.11 kW for PV, 10 kW for the biogas generator, 5 batteries and 3.96 kW converter. Another important remarque is that the highest amount of electricity consumption in the village occurs mainly during nighttime which explains why the biogas generator has a higher capacity than the PV array.

The COE of this system was found to be equal to 0.362 US\$, which is relatively close to the COE in the current study. This further confirms the previous conclusion that the use of batteries significantly increases the HRES cost. It is worth mentioning that many studies of various HRES found similar conclusions for instance, we cite^{57,58}

Although the HRES design and costs depend on many parameters such as the renewable energy source availability, the load profile, demand-side management techniques, and the fluctuation of the components' price.⁵⁹ Nevertheless, the HRES

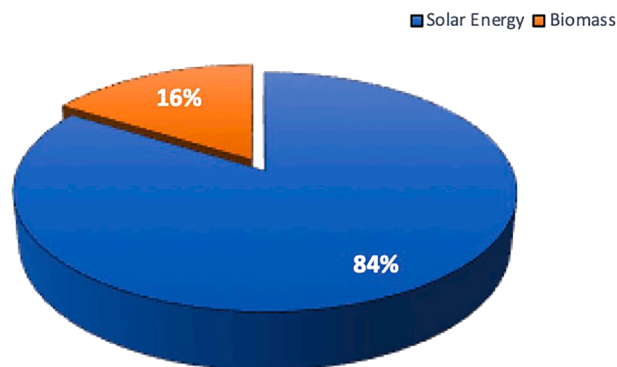


Figure 8. Distribution of Energy Sources for Scenario 4

proposed in this study has shown a great reliability, flexibility, and cost-effectiveness, especially for load profiles with a period of high peaks, particularly in summertime, and a tremendous potential for CO₂ emissions reduction; therefore, it can be adapted and scaled according to other locations' renewable energy sources and load profiles.

Sensitivity analysis

In this section, sensitivity analysis is explored. To this end, we will explore the impact of key financial parameters on the technical and economic performance of the optimal combination, which includes photovoltaic energy, biomass, and battery systems. To achieve this, we varied the discount rate and inflation rate to assess their effects on the cost of energy (COE), a vital techno-economic indicator. Initially, we varied the discount rate while maintaining a constant inflation rate of 2.5%. Subsequently, we repeated the process by adjusting the inflation rate while keeping the discount rate fixed at 5%. In both scenarios, the variations were implemented in increments of 0.5%, ranging from 0% to 10%.

Figure 13 illustrates the impact of varying both the discount rate and inflation rate on the Cost of Energy (COE) for the optimal system. The graphical representation indicates that an increase in the discount rate results in a corresponding rise in the COE. Conversely, an increase in the inflation rate leads to a decrease

in the COE. Specifically, at a 0% discount rate, a COE of 0.244 \$/kWh can be achieved, while at a 10% discount rate, the COE increases to 0.42 \$/kWh. In terms of inflation, a 0% rate results in a COE of 0.378 \$/kWh, whereas at 10%, the COE drops to 0.208 \$/kWh. These findings highlight the significant influence of the discount rate and inflation rate on system costs, thereby affecting its overall economic viability.

This study finding provides an important insight for the role that various renewable energy sources can play in HRES for power generation, which can help developing countries such as Morocco achieve their goals in terms of energy security, economic growth, and sustainability. Indeed, HRES has several advantages in terms of improving energy access and reliability, reducing reliance on energy imports by using utilizing local renewable energy resources, and reducing carbon emissions.

Moreover, HRES have the great advantage of being scalable as they can start small and scale up based on demand to power small villages, industries, or entire cities while being adapted to the local resources based on availability. In fact, the proposed HRES can be adapted for larger applications and in different contexts, for example, the use of excess electricity to generate green hydrogen, which can be used as a storage medium, which can help in the case of peak demand, as observed in this study. Also, the possibility of grid connection and smart grid integration can ensure power stability, optimizing energy distribution and reduce reliance on fossil fuels. In addition, such a system may be ideal for factories and industrial applications. In this context, HRES can be a part of microgrids in industrial zones, which can help cut costs and emissions. Besides, biomass and industrial waste can be utilized for power generation. Moreover, the HRES can be adapted to be an integral part of Urban, Smart City applications such as for combined heat, and power as the biomass can be used for heat generation. It can also help with the integration of electric vehicles as they can serve as mobile batteries for the HRES networks, which can revolutionize public transport and solve issues for electric vehicle charging infrastructure.

However, the scaling of the HRES at the national level will require an appropriate policy framework at the national level. The government should set renewable energy targets in hybrid grids to ensure a high share of renewable energies, which will

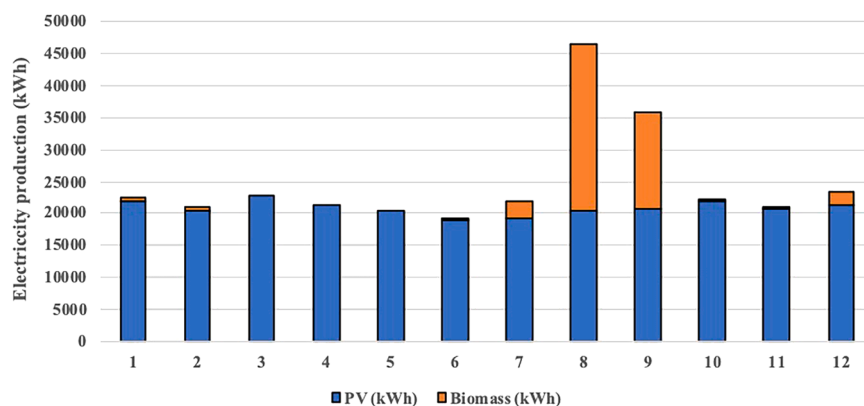


Figure 9. Monthly electrical power generation

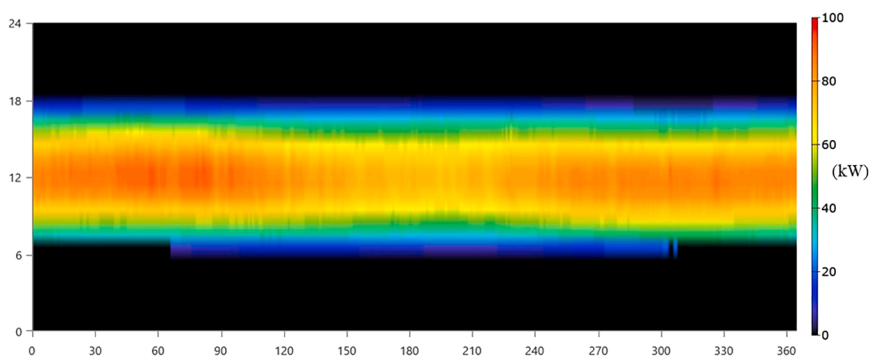


Figure 10. Power generation from the PV array

Practical implementations and future works

The results of this study provide important insights into the role of renewable energy integration in achieving self-sufficiency in electricity demand for governmental buildings -specifically, research centers such as the one examined in this case study. This is particularly relevant

for countries with abundant renewable resources, such as Morocco, and can contribute significantly to national goals related to energy security, economic development, and environmental sustainability. Hybrid Renewable Energy Systems (HRES) offer several advantages, including enhanced energy access and reliability, reduced dependence on energy imports through the utilization of local renewable resources, and lower carbon emissions. Moreover, HRES have a high potential for practical implementation with the advantage of being scalable as they can be adapted for a single building or scale up to meet the demand of small villages, industries, or entire cities while being adapted to the local resources based on availability. In fact, the proposed HRES can be adapted for larger applications and in different contexts, for example, the use of excess electricity to generate green hydrogen, which can be, used as a storage medium, which can help in the case of peak demand as observe in this study. Also, the possibility of grid connection and smart grid integration can ensure power stability, optimizing energy distribution, and reduce reliance on fossil fuels. In addition, such a system may be ideal for factories and industrial applications. In this context, HRES can be a part of microgrids in industrial zones, which can help cut costs and emissions. Besides, biomass and industrial waste can be utilized for power generation. Moreover, the HRES can be adapted to be an integral part of Urban, Smart City applications such as for combined heat, and power as the biomass can be used for heat generation. It can also help with the integration of electric vehicle as they can serve as mobile batteries for the HRES networks, which can revolutionize public transport and solving issues for electric vehicle charging infrastructure.

require support for energy storage solutions such as hydrogen to balance intermittent renewables and the adoption of smart grid technology to optimize energy fraction between renewable and non-renewable energy sources. Also, the government needs to provide financial incentive schemes to adopt hybrid energy solutions and, more importantly, the adoption of net metering policies to allow excess renewable energy from HRES to be sold back into the grid. On the other hand, the implementation of the carbon tax for intensive energy sources while providing tax breaks or subsidies for HRES can drive the investment toward cleaner technologies. In addition, the grid infrastructure should be optimized and transmission networks upgraded to efficiently integrate multiple energy sources. Moreover, the promotion of research, development, and innovation on HRES by funding research for next generation HRES driven by artificial intelligence energy management and new storage technologies and strengthen international collaboration to share best practices across countries and to develop scalable hybrid solutions.

In future work, the use of hydrogen based fuel cell and other battery technologies will be explored, as well as the modeling and optimization of the proposed HRES under different climatic conditions and biomass sources. In addition, the implications of the proposed policies on the financial and economic competitiveness will be explored. Another aspect that we are expecting to implement is the impact of soiling on the HRES system performance. Indeed, based on already conducted experiments,^{49,60,61} the impact of dust deposition and mitigation/cleaning strategies will be implemented on the simulated models to enhance the performance of the proposed strategies by optimizing cleaning costs and the LCOE.

valant for countries with abundant renewable resources, such as Morocco, and can contribute significantly to national goals related to energy security, economic development, and environmental sustainability. Hybrid Renewable Energy Systems (HRES) offer several advantages, including enhanced energy access and reliability, reduced dependence on energy imports through the utilization of local renewable resources, and lower carbon emissions.

Moreover, HRES have a high potential for practical implementation with the advantage of being scalable as they can be adapted for a single building or scale up to meet the demand of small villages, industries, or entire cities while being adapted to the local resources based on availability. In fact, the proposed HRES can be adapted for larger applications and in different contexts, for example, the use of excess electricity to generate green hydrogen, which can be, used as a storage medium, which can help in the case of peak demand as observe in this study. Also, the possibility of grid connection and smart grid integration can ensure power stability, optimizing energy distribution, and reduce reliance on fossil fuels. In addition, such a system may be ideal for factories and industrial applications. In this context, HRES can be a part of microgrids in industrial zones, which can help cut costs and emissions. Besides, biomass and industrial waste can be utilized for power generation. Moreover, the HRES can be adapted to be an integral part of Urban, Smart City applications such as for combined heat, and power as the biomass can be used for heat generation. It can also help with the integration of electric vehicle as they can serve as mobile batteries for the HRES networks, which can revolutionize public transport and solving issues for electric vehicle charging infrastructure.

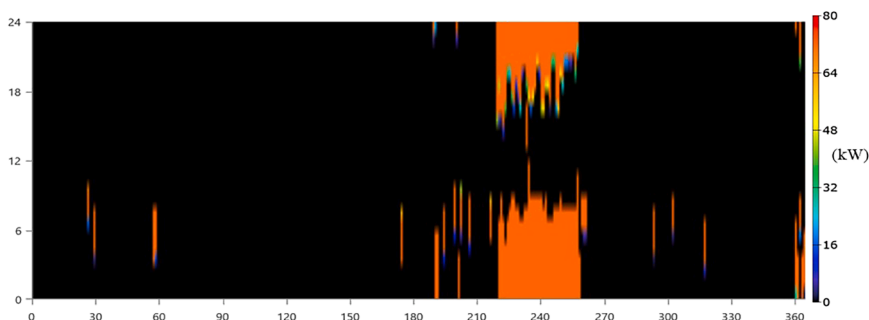


Figure 11. Power generation from the biogas generator

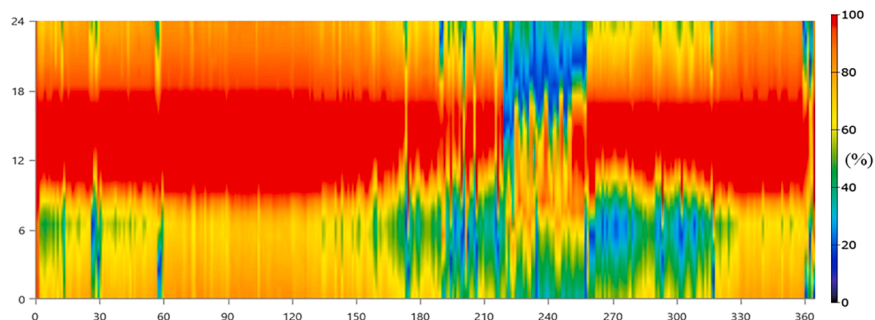


Figure 12. Batteries state of charge

In future work, the climate impact will be considered, as well as the use of hydrogen based fuel cell and other battery technologies will be explored, as well as the modeling and optimization of the proposed HRES under different climatic conditions and biomass sources. In addition, the implications of the proposed policies on the financial and economic competitiveness will be explored.

Conclusion

Due to the massive growth of the industrial sector in Morocco, the growth of energy demand became more and more difficult to deal with, especially with the increase in gas and oil prices due to the COVID-19 pandemic and the Russian-Ukrainian conflict, and the lack of fossil resources in Morocco. On the other hand Morocco has a great potential of renewable sources -mainly solar and wind-that can be used to overcome the energy dependence and may lead to a 100% renewable integration, especially on governmental organizations, which contribute to the shift of a green economy and a sustainable development. To this end, the current study aims to identify the optimal design and performance of hybrid renewable energy systems (HRES) for meeting the entire energy demand of the Green Energy Park (GEP), a research facility located in Benguerir, Morocco. Using high-resolution local meteorological data, the HOMER software was employed to model and evaluate different combinations of renewable technologies, such as photovoltaic (PV) solar panels, wind turbines, biomass generators, and battery storage. Nine distinct configurations were assessed to determine the most cost-effective and technically viable solution for achieving a 100% renewable energy supply for the institute. The key results of this study reveal that.

- (1) By analyzing the different scenarios, it becomes evident that PV consistently contributed the largest share of en-

ergy production in most configurations, particularly in those with high solar availability, where it accounted for up to 97% of the total energy generated in PV/Wind/Battery systems. In contrast, wind energy made only a minimal contribution due to the limited wind resources at the site. Biomass played a critical role during high-demand months, especially in August and September, compensating for the reduced efficiency of PV systems during the hot, dusty summer months.

- (2) For this case study, the most optimal and reliable hybrid renewable energy system is the PV/Biomass/Battery combination, with solar energy contributing 83% and biomass providing 16.2%. This configuration offers the lowest cost of energy (COE) at \$0.333/kWh, along with a substantial energy surplus of up to 35%, which can be stored in batteries. This surplus storage significantly reduces operational costs by decreasing reliance on biomass during peak demand periods. The excess solar energy can be stored for use during nighttime or cloudy days and could potentially be used for hydrogen production or water desalination.
- (3) Regarding the sensitivity analysis, it was observed that the COE increased as the discount rate rose, reaching \$0.42/kWh at a 10% discount rate. Conversely, the COE decreased with higher inflation rates, dropping to \$0.208/kWh at a 10% inflation rate. These results highlight the system's significant sensitivity to financial parameters, emphasizing the impact of economic conditions on the overall cost-effectiveness of the hybrid renewable energy system.

Table 11. Overview of diverse expenses associated with the PV/Biomass/Battery hybrid system

Component	Capital (\$)	Replacement (\$)	O&M (\$)	Total (\$)
Battery	122 430.00	85 291.62	58 170.32	215 485.27
Biogas Generator	120 000	0	81 493.32	395 026.20
PV	187 574.17	0	36 762.97	224 337.14
Converter	41 511.41	28 919.18	0	53 339.59
System	471 515.59	114 210.80	176 426.61	888 188.20

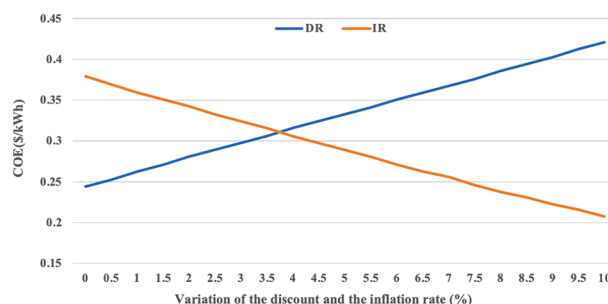


Figure 13. Impact of Variation in Discount Rate (DR) and Inflation Rate (IR) on Electricity Cost in the Optimal System

These results hold significant importance, providing valuable insights for the Moroccan government and policymakers regarding the integration of renewable energy within governmental organizations. Such investments are crucial for fostering green and sustainable development and decreasing the energy bill and dependence of the country.

Limitations of the study

Authors would like to mention that this study has a limitation as it does not consider the impact of harsh climate conditions for semi-arid regions like Morocco, such as soiling and high temperature values that may affect the production and the durability of the HRES system Components. Indeed, based on already conducted experiments,^{49,60,61} the impact of dust deposition and mitigation/cleaning strategies will be implemented on the simulated models to enhance the performance of the proposed strategies by optimizing cleaning costs and the LCOE.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Prof. Dr Ahmed ALAMI MERROUNI (alami.univ.oujda@gmail.com).

Materials availability

This study did not generate new unique reagents.

Data and code availability

- All original data will be made available from the [lead contact](#) upon request.
- All original code will be made available from the [lead contact](#) upon request.
- Any additional information required to analyze the data reported in this article is available from the [lead contact](#) upon request.

ACKNOWLEDGMENTS

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AUTHOR CONTRIBUTIONS

Conceptualization, NE.C, S.T, and A.A; methodology, A.A.M; T.S; software, NE.C, S.T, and A.A; writing – original draft, NE.C, S.T, and A.A; writing – review and editing, NE.C, S.T, and A.A and A.A.M; visualization, A.A.M.; supervision, A.A.M.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- [KEY RESOURCES TABLE](#)
- [METHOD DETAILS](#)
- [QUANTIFICATION AND STATISTICAL ANALYSIS](#)
 - Homer-pro software description
 - Greenius software description

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REFERENCES

1. Elfeky, K.E., and Wang, Q. (2023). Techno-economic assessment and optimization of the performance of solar power tower plant in Egypt's climate conditions. *Energy Convers. Manag.* 280, 116829. <https://doi.org/10.1016/j.enconman.2023.116829>.
2. Tahir, S., Ahmad, M., Abd-ur-Rehman, H.M., and Shakir, S. (2021). Techno-economic assessment of concentrated solar thermal power generation and potential barriers in its deployment in Pakistan. *J. Clean. Prod.* 293, 126125. <https://doi.org/10.1016/j.jclepro.2021.126125>.
3. Ingenhoven, P., Lee, L., Saw, W., Rafique, M.M., Potter, D., and Nathan, G.J. (2023). Techno-economic assessment from a transient simulation of a concentrated solar thermal plant to deliver high-temperature industrial process heat. *Renew. Sustain. Energy Rev.* 185, 113626. <https://doi.org/10.1016/j.rser.2023.113626>.
4. Aseri, T.K., Sharma, C., and Kandpal, T.C. (2021). Estimation of capital costs and techno-economic appraisal of parabolic trough solar collector and solar power tower based CSP plants in India for different condenser cooling options. *Renew. Energy* 178, 344–362. <https://doi.org/10.1016/j.renene.2021.05.166>.
5. Aseri, T.K., Sharma, C., and Kandpal, T.C. (2022). A techno-economic appraisal of parabolic trough collector and central tower receiver based solar thermal power plants in India: Effect of nominal capacity and hours of thermal energy storage. *J. Energy Storage* 48, 103976. <https://doi.org/10.1016/j.est.2022.103976>.
6. Gutiérrez, R.E., Haro, P., and Gómez-Barea, A. (2021). Techno-economic and operational assessment of concentrated solar power plants with a dual supporting system. *Appl. Energy* 302, 117600. <https://doi.org/10.1016/j.apenergy.2021.117600>.
7. Rad, M.A.V., Ghasempour, R., Rahdan, P., Mousavi, S., Arastounia, M., and Arastounia, M. (2020). Techno-economic analysis of a hybrid power system based on the cost-effective hydrogen production method for rural electrification, A case study in Iran. *Energy* 190, 116421. <https://doi.org/10.1016/j.energy.2019.116421>.
8. Odou, O.D.T., Bhandari, R., Adamou, R., and Adamou, R. (2020). Hybrid off-grid renewable power system for sustainable rural electrification in Benin. *Renew. Energy* 145, 1266–1279. <https://doi.org/10.1016/j.renene.2019.06.032>.
9. Vivas, F.J., De las Heras, A., Segura, F., and Andújar, J.M. (2018). A review of energy management strategies for renewable hybrid energy systems with hydrogen backup. *Renew. Sustain. Energy Rev.* 82, 126–155. <https://doi.org/10.1016/j.rser.2017.09.014>.
10. Okundamiya, M.S. (2021). Size optimization of a hybrid photovoltaic/fuel cell grid connected power system including hydrogen storage. *Int. J. Hydrogen Energy* 46, 30539–30546. <https://doi.org/10.1016/j.ijhydene.2020.11.185>.
11. Al-Sharafi, A., Sahin, A.Z., Ayar, T., and Yilbas, B.S. (2017). Techno-economic analysis and optimization of solar and wind energy systems for power generation and hydrogen production in Saudi Arabia. *Renew. Sustain. Energy Rev.* 69, 33–49. <https://doi.org/10.1016/j.rser.2016.11.157>.
12. Suresh, V., Muralidhar, M., and Kiranmayi, R. (2020). Modelling and optimization of an off-grid hybrid renewable energy system for electrification in a rural areas. *Energy Rep.* 6, 594–604. <https://doi.org/10.1016/j.egyr.2020.01.013>.
13. Pal, P., and Mukherjee, V. (2021). Off-grid solar photovoltaic/hydrogen fuel cell system for renewable energy generation: An investigation based on techno-economic feasibility assessment for the application of end-user load demand in North-East India. *Renew. Sustain. Energy Rev.* 149, 111421. <https://doi.org/10.1016/j.rser.2021.111421>.

14. Al-Ghussain, L., Darwish Ahmad, A., Abubaker, A.M., and Mohamed, M.A. (2021). An integrated photovoltaic/wind/biomass and hybrid energy storage systems towards 100% renewable energy microgrids in university campuses. *Sustain. Energy Technol. Assess.* 46, 101273. <https://doi.org/10.1016/j.seta.2021.101273>.
15. Ribó-Pérez, D., Herraiz-Cañete, Á., Alfonso-Solar, D., Vargas-Salgado, C., and Gómez-Navarro, T. (2021). Modelling biomass gasifiers in hybrid renewable energy microgrids; a complete procedure for enabling gasifiers simulation in HOMER. *Renew. Energy* 174, 501–512. <https://doi.org/10.1016/j.renene.2021.04.083>.
16. Guerello, A., Page, S., Holburn, G., and Balzarova, M. (2020). Energy for off-grid homes: Reducing costs through joint hybrid system and energy efficiency optimization. *Energy Build.* 207, 109478. <https://doi.org/10.1016/j.enbuild.2019.109478>.
17. Bahramara, S., Moghaddam, M.P., and Haghighi, M.R. (2016). Optimal planning of hybrid renewable energy systems using HOMER: A review. *Renew. Sustain. Energy Rev.* 62, 609–620. <https://doi.org/10.1016/j.rser.2016.05.039>.
18. Li, C., Zhou, D., Zhang, L., and Shan, Y. (2022). Exploration on the feasibility of hybrid renewable energy generation in resource-based areas of China: Case study of a regeneration city. *Energy Strateg. Rev.* 42, 100869. <https://doi.org/10.1016/j.esr.2022.100869>.
19. Tiam, K.P., Medjo Nouadje, B.A., Chegnimonhan, V., Tchien, G., and Tchinda, R. (2022). Techno-economic feasibility of a PV/battery/fuel cell/electrolyzer/biogas hybrid system for energy and hydrogen production in the far north region of cameroon by using HOMER pro. *Energy Strateg. Rev.* 44, 100988. <https://doi.org/10.1016/j.esr.2022.100988>.
20. Turkdogan, S. (2021). Design and optimization of a solely renewable based hybrid energy system for residential electrical load and fuel cell electric vehicle. *Eng. Sci. Technol. An. Int. J.* 24, 397–404. <https://doi.org/10.1016/j.jestch.2020.08.017>.
21. Chaichan, W., Waewsak, J., Nikhom, R., Kongruang, C., Chiwamongkhonkarn, S., and Gagnon, Y. (2022). Optimization of stand-alone and grid-connected hybrid solar/wind/fuel cell power generation for green islands: Application to Koh Samui, southern Thailand. *Energy Rep.* 8, 480–493. <https://doi.org/10.1016/j.egyr.2022.07.024>.
22. Odeto, O.A., Olulope, P.K., Olanrewaju, O.M., Alimi, A.O., and Igbinosa, O.G. (2023). Multi-year techno-economic assessment of proposed zero-emission hybrid community microgrid in Nigeria using HOMER. *Heliyon* 9, e19189. <https://doi.org/10.1016/j.heliyon.2023.e19189>.
23. Chisale, S.W., Eliya, S., and Taalo, J. (2023). Optimization and design of hybrid power system using HOMER pro and integrated CRITIC-PROMETHEE II approaches. *Green Technol. Sustain.* 1, 100005. <https://doi.org/10.1016/j.grets.2022.100005>.
24. Baneshi, M., and Hadianfard, F. (2016). Techno-economic feasibility of hybrid diesel/PV/wind/battery electricity generation systems for non-residential large electricity consumers under southern Iran climate conditions. *Energy Convers. Manag.* 127, 233–244. <https://doi.org/10.1016/j.enconman.2016.09.008>.
25. Li, C., Zhou, D., Wang, H., Lu, Y., and Li, D. (2020). Techno-economic performance study of stand-alone wind/diesel/battery hybrid system with different battery technologies in the cold region of China. *Energy* 192, 116702. <https://doi.org/10.1016/j.energy.2019.116702>.
26. Allouhi, A., Zamzoum, O., Islam, M.R., Saidur, R., Kousksou, T., Jamil, A., and Derouich, A. (2017). Evaluation of wind energy potential in Morocco's coastal regions. *Renew. Sustain. Energy Rev.* 72, 311–324. <https://doi.org/10.1016/j.rser.2017.01.047>.
27. Ayodele, T.R., Ogunjuyigbe, A.S.O., and Amusan, T.O. (2016). Wind Power Utilization Assessment and Economic Analysis of Wind Turbines across Fifteen Locations in the Six Geographical Zones of Nigeria. *J. Clean. Prod.* 129, 341–349. <https://doi.org/10.1016/j.jclepro.2016.04.060>.
28. Zahboune, H., Zouggar, S., Krajacic, G., Varbanov, P.S., Elhafyani, M., and Ziani, E. (2016). Optimal hybrid renewable energy design in autonomous system using Modified Electric System Cascade Analysis and Hom-er software. *Energy Convers. Manag.* 126, 909–922. <https://doi.org/10.1016/j.enconman.2016.08.061>.
29. Das, B.K., Hoque, N., Mandal, S., Pal, T.K., and Raihan, M.A. (2017). A techno-economic feasibility of a stand-alone hybrid power generation for remote area application in Bangladesh. *Energy* 134, 775–788. <https://doi.org/10.1016/j.energy.2017.06.024>.
30. Rajbongshi, R., Borgohain, D., and Mahapatra, S. (2017). Optimization of PV-biomass-diesel and grid base hybrid energy systems for rural electrification by using HOMER. *Energy* 126, 461–474. <https://doi.org/10.1016/j.energy.2017.03.056>.
31. Duman, A.C., and Güler, Ö. (2018). Techno-economic analysis of off-grid PV/wind/fuel cell hybrid system combinations with a comparison of regularly and seasonally occupied households. *Sustain. Cities Soc.* 42, 107–126. <https://doi.org/10.1016/j.scs.2018.06.029>.
32. Das, H.S., Tan, C.W., Yatim, A.H.M., and Lau, K.Y. (2017). Feasibility analysis of hybrid photovoltaic/battery/fuel cell energy system for an indigenous residence in East Malaysia. *Renew. Sustain. Energy Rev.* 76, 1332–1347. <https://doi.org/10.1016/j.rser.2017.01.174>.
33. Salisu, S., Mustafa, M.W., Olatomiwa, L., and Mohammed, O.O. (2019). Assessment of technical and economic feasibility for a hybrid PV-wind-diesel-battery energy system in a remote community of north central Nigeria. *Alexandria Eng. J.* 58, 1103–1118. <https://doi.org/10.1016/j.aej.2019.09.013>.
34. Hafez, O., and Bhattacharya, K. (2012). Optimal planning and design of a renewable energy based supply system for microgrids. *Renew. Energy* 45, 7–15. <https://doi.org/10.1016/j.renene.2012.01.087>.
35. Ajayi, O.O., Ohijeagbon, O.D., Mercy, O., and Ameh, A. (2016). Potential and econometrics analysis of standalone RE facility for rural community utilization and embedded generation in North-East, Nigeria. *Sustain. Cities Soc.* 21, 66–77. <https://doi.org/10.1016/j.scs.2016.01.003>.
36. Kalinci, Y., Hepbasli, A., and Dincer, I. (2015). Techno-economic analysis of a stand-alone hybrid renewable energy system with hydrogen production and storage options. *Int. J. Hydrogen Energy* 40, 7652–7664. <https://doi.org/10.1016/j.ijhydene.2014.10.147>.
37. Boussetta, M., El Bachtiri, R., Khanfara, M., and El Hammoui, K. (2017). Assessing the potential of hybrid PV-Wind systems to cover public facilities loads under different Moroccan climate conditions. *Sustain. Energy Technol. Assess.* 22, 74–82. <https://doi.org/10.1016/j.seta.2017.07.005>.
38. El-houari, H., Allouhi, A., Rehman, S., Buker, M.S., Kousksou, T., Jamil, A., and El Amrani, B. (2020). Feasibility evaluation of a hybrid renewable power generation system for sustainable electricity supply in a Moroccan remote site. *J. Clean. Prod.* 277, 123534. <https://doi.org/10.1016/j.jclepro.2020.123534>.
39. Singh, A., and Baredar, P. (2016). Techno-economic assessment of a solar PV, fuel cell, and biomass gasifier hybrid energy system. *Energy Rep.* 2, 254–260. <https://doi.org/10.1016/j.egyr.2016.10.001>.
40. Babu, M.K., and Ray, P. (2023). Sensitivity analysis, optimal design, cost and energy efficiency study of a hybrid forecast model using HOMER pro. *J. Eng. Res.* 11, 100033. <https://doi.org/10.1016/j.jer.2023.100033>.
41. Kaur Channi, H., Singh, M., Singh Brar, Y., Dhir, A., Gupta, S., Singh, H., Kumar, R., and Kaur, S. (2022). Agricultural waste assessment for the optimal power generation in the Ludhiana district, Punjab, India. *Mater. Today Proc.* 50, 700–708.
42. Mahdavi, M., and Vera, D. (2023). Importance of renewable energy sources and agricultural biomass in providing primary energy demand for Morocco. *Int. J. Hydrogen Energy* 48, 34575–34598.
43. Mouaky, A., and Racheq, A. (2020). Energetic, exergetic and exergoeconomic assessment of a hybrid solar/biomass polygeneration system: A case study of a rural community in a semi-arid climate. *Renew. Energy* 158, 280–296. <https://doi.org/10.1016/j.renene.2020.05.135>.
44. Mouaky, A., and Racheq, A. (2020). Thermodynamic and thermo-economic assessment of a hybrid solar/biomass polygeneration system under

- the semi-arid climate conditions. *Renew. Energy* 156, 14–30. <https://doi.org/10.1016/j.renene.2020.04.019>.
45. Touili, S., Bouaichi, A., Alami Merrouni, A., Amrani, A.-illah, El Amrani, A., El Hassouani, Y., and Messaoudi, C. (2022). Performance analysis and economic competitiveness of 3 different PV technologies for hydrogen production under the impact of arid climatic conditions of Morocco. *Int. J. Hydrogen Energy* 47, 31596–31613.
 46. Mehdi, M., Ammari, N., Alami Merrouni, A., Benazzouz, A., and Dahmani, M. (2023). Experimental investigation on the effect of wind as a natural cooling agent for photovoltaic power plants in desert locations. *Case Stud. Therm. Eng.* 47, 103038. <https://doi.org/10.1016/j.renene.2024.120167>.
 47. Ouali, H.A.L., Ahmed, A.M., Amine Moussaoui, M., and Ahmed, M. (2015). Electricity yield analysis of a 50 MW solar tower plant under Moroccan climate. In *2015 International Conference on Electrical and Information Technologies (ICEIT) (IEEE)*, pp. 252–256.
 48. Mehdi, M., Ammari, N., Alami Merrouni, A., Elhamaoui, S., and Dahmani, M. (2024). Innovative design and field performance evaluation of a desert-adapted PV module for enhanced solar energy harvesting and reliability in harsh arid environments. *Appl. Energy* 366, 123359.
 49. Ammari, N., Mehdi, M., Alami Merrouni, A., Benazzouz, A., and Chaabelasri, E. (2024). In-situ soiling evaluation and cleaning schedules optimization for several PV technologies under desert climate. *Renew. Energy* 224, 120167.
 50. Ammari, N., Mehdi, M., Alami Merrouni, A., and Benazzouz, A. (2023). Experimental analysis of Anti-Reflective coating performance in desert Climate: Yield analysis, soiling impact and cleaning durability evaluation. *Sustain. Energy Technol. Assess.* 60, 103547.
 51. Mehdi, M., Ammari, N., Alami Merrouni, A., El Gallassi, H., Dahmani, M., and Ghennioui, A. (2023). An experimental comparative analysis of different PV technologies performance including the influence of hot-arid climatic parameters: Toward a realistic yield assessment for desert locations. *Renew. Energy* 205, 695–716.
 52. Ammari, N., Alami Merrouni, A., Mendyl, A., Chaabelasri, E., and Weidinger, T. (2024). Energy, Economic and Environmental (3E) Analysis for an Optimal CSP Technology Integration in Morocco. *Energies* 17, 3020.
 53. Allouhi, A., Saadani, R., Buker, M.S., Kousksou, T., Jamil, A., and Rahmoune, M. (2019). Energetic, economic and environmental (3E) analyses and LCOE estimation of three technologies of PV grid-connected systems under different climates. *Sol. Energy* 178, 25–36. <https://doi.org/10.1016/j.solener.2018.11.060>.
 54. Lund, C., and Biswas, W. (2008). A review of the application of lifecycle analysis to renewable energy systems. *Bull. Sci. Technol. Soc.* 28, 200–209.
 55. Kumar, P., Channi, H.K., Kumar, R., Prakash, C., Bhowmik, A., Pandey, S., Singh, A.K., Ali, M.M., and Soudagar, M.E.M. (2024). Efficient integration of photo voltaic and hydro energy technologies for sustainable power generation in rural areas: A case study. *Mater. Sci. Energy Technol.* 7, 297–308.
 56. Kumar, R., and Channi, H.K. (2022). A PV-Biomass off-grid hybrid renewable energy system (HRES) for rural electrification: Design, optimization and techno-economic-environmental analysis. *J. Clean. Prod.* 349, 131347. <https://doi.org/10.1016/j.jclepro.2022.131347>.
 57. Kaur, G., Kaur, J., Sharma, A., Jain, A., Kumar, R., Kumar, R., Alsubih, M., Islam, S., and Ammarullah, M.I. (2024). A Techno-economic investigation and empowering rural resilience through bioengineering: a case study on self-sustainable village energy models. *Int. J. Low Carbon Technol.* 19, 1275–1287. <https://doi.org/10.1093/ijlct/ctae072>.
 58. Kaur, H., Kumar, R., Kumar, P., Kaur, S., Dixit, K.K., and Sharma, H. (2024). Efficient and Cost-Effective Renewable Energy Integration of Photovoltaic and Hydro in Rural India Using HOMER Pro: A Case Study of Chupki, Punjab. In *Intelligent Manufacturing and Energy Sustainability. ICIMES 2023. Smart Innovation, Systems and Technologies*, 372, P.H.V.S. Talpa Sai, S. Potnuru, M. Avcar, and V. Ranjan Kar, eds. (Springer). https://doi.org/10.1007/978-981-99-6774-2_26.
 59. Channi, H.K., and Kumar, R. (2022). The Role of Smart Sensors in Smart City. In *Smart Sensor Networks. Studies in Big Data*, 92, U. Singh, A. Abraham, A. Kaklauskas, and T.P. Hong, eds. (Springer), pp. 27–48. https://doi.org/10.1007/978-3-030-77214-7_2.
 60. Merrouni, A.A., Conceição, R., Mouaky, A., Silva, H.G., and Ghennioui, A. (2020). CSP performance and yield analysis including soiling measurements for Morocco and Portugal. *Renew. Energy* 162, 1777–1792. <https://doi.org/10.1016/j.renene.2020.10.014>.
 61. Mehdi, M., Conceição, R., Ammari, N., Ahmed, A.M., González-Aguilar, J., and Mohamed, D. (2024). Modeling, assessment and characterization of soiling on PV Technologies. Toward a Better understanding of the Relation between dust deposition and performance losses. *Sustain. Energy Technol. Assess.* 71, 104023. <https://doi.org/10.1016/j.seta.2024.104023>.

STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Software and algorithms		
GREENIUS SOFTWARE- FREE OPEN ACCESS	https://www.dlr.de/en/sf/research-and-transfer/research-services/simulation-and-profitability-assessment/greenius-software-tool	DLR (CC BY-NC-ND 3.0)
HomerPro - Students free trial version	http://homerenergy.com	HomerPRO

METHOD DETAILS

This study presents a simulation-based case analysis aimed at designing a 100% renewable hybrid energy system to meet the energy demands of the Green Energy Park, a research facility located in Benguerir, in the mid-southern region of Morocco.

The first step involved gathering the necessary input data, which included: (i) The electricity consumption profile (used to define the energy demand), and (ii) Local meteorological data-solar irradiation, ambient temperature, humidity, pressure, wind speed and direction -to simulate the energy output from various renewable sources. A thorough quality check of the input data was performed to ensure its accuracy, particularly for the meteorological data, which was accurately analyzed to correct and missing point, non-logical measurement or physical limit.

The next step consisted of identifying the renewable energy technologies to be used and defining the simulation scenarios. Four technologies were selected: photovoltaic (PV), concentrated solar power (CSP), wind, and biomass, along with a battery storage system. These technologies were combined into nine different configurations, designed based on insights from existing literature and the diversity of sources.

Each scenario was then modeled and simulated using the HOMER software. However, since the used version of HOMER -Students free trial version-does not support direct simulation of CSP technologies, the open-access software Greenius was employed to simulate CSP output based on the meteorological data. The resulting CSP production profiles were then integrated into the HOMER simulations. Simultaneously, an optimization analysis - focusing on the lowest electricity costs while ensuring a 100% renewable supply- was carried out using HOMER for the whole scenarios.

Following the identification of the optimal hybrid renewable energy system (HRES), the associated CO₂ emissions reductions were calculated for each component and compared with those from similar studies conducted in comparable climatic regions.

Finally, a sensitivity analysis was performed to assess the impact of key economic parameters -namely, the discount rate and inflation rate-on the cost of electricity (COE). This involved varying these values to evaluate their influence on the overall system performance and economics.

QUANTIFICATION AND STATISTICAL ANALYSIS

In this study no statistical analysis has been conducted. The errors appearing on the section [renewable resources assessment](#) are related to the relative errors of the used instruments which are provided by the manufacturer.

Homer-pro software description

The HOMER Pro software (Hybrid Optimization of Multiple Energy Resources), developed by the U.S. National Renewable Energy Laboratory (NREL), is a powerful and widely used tool for simulating and optimizing hybrid renewable energy systems. This tool allows users to design and evaluate various configurations of such systems. To run the simulation, inputs such as the electrical load profile and meteorological data must be uploaded. To supply the load, users define one or more power sources. HOMER supports a wide range of renewable energy technologies, including photovoltaic (PV) systems, wind turbines, biomass generators, hydropower, and battery storage systems. Besides, this tool enables the importation of yield from other technologies that are not supported -like CSP.

With its user-friendly interface, users can freely select any combination of these technologies depending on their study objectives. For economic analysis, users must also enter the relevant financial parameters for each component, including capital cost, replacement cost, operation and maintenance costs, discount rate, and project lifetime. These inputs are essential for HOMER to conduct a reliable techno-economic analysis, enabling it to simulate system behavior over time and compute long-term financial metrics. Once all inputs are provided, HOMER automatically generates a large number of system configurations, combining the selected

components into different scenarios. The tool then performs an optimization analysis based on the Total Net Present Cost (NPC) and the Levelized Cost of Energy (LCOE), ranking the configurations from most to least cost-effective. These ranking results provide users with strong decision-making support in selecting the optimal renewable energy configuration for the case under study.

Greenius software description

Greenius (Green Energy System Analysis Tool) is a free and powerful software developed by the DLR (German Aerospace Center) to simulate the yield of various renewable energy systems. To run a simulation, the software requires a meteorological file for the selected location and details of the power plant configuration, such as the number of collectors, type of working fluid, power block capacity, and storage options-in the case of CSP power plants. The simulation results provide a wide range of outputs related to plant operation, including electricity production, heat generation, storage charge/discharge rates, system efficiency, and more. In this study, Greenius was used to simulate the electrical output of the CSP (Concentrated Solar Power) plant, which was then imported into HOMER Pro, as this tool does not support CSP as an energy system.